Simulation and Analysis of BB84 Quantum Key Distribution for Secure Data Transmission

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***Abstract*— With the fast growth of computational technologies, data security has played a significant role in modern communication systems. Traditional cryptographic techniques depend on hard mathematical issues that are likely to be compromised by quantum computer advancements. Quantum Key Distribution (QKD) provides a secure alternative as it does not rely on computational assumptions but rather yields security based on the laws of quantum physics. In this paper, we focus on understanding QKD more in-depth with a simulation-based study of one of the most adopted methods: BB84 protocol. The protocol is implemented using Python to observe its behavior under the presence of various noise levels and an eavesdropper. Metrics like QBER and secret key generation rate are used to measure the system performance. It shows that as the noise in the channel increases; the error also increases. Moreover, the inclusion of eavesdropper adds a disturbance to this state, resulting in high level of QBER. Simultaneously, the key generation rate is lowered as the error rate increases demonstrating a trade-off between security and speed of working. They showcase the BB84 protocol's ability to detect eavesdropping and enable acceptance of correction measures.**

Keywords*— *Quantum Key Distribution, BB84 Protocol, Quantum Cryptography, Quantum Bit Error Rate, Secure Communication

I. Introduction

Now that digital communication systems are increasingly being used, securing sensitive information has emerged as a key challenge. RSA and elliptic curve cryptography are classic symmetric crypto systems based on the computational hardness of neat math problems. The arrival of quantum computing, however, threatens these methods quite severely, as quantum algorithms may be able to solve such problems more efficiently. To deal with this problem, quantum cryptography as a new approach has been offered that is used based on quantum mechanics laws. Quantum Key Distribution (QKD) enables two parties to produce a shared secret key in a manner such that it would be detectable if an eavesdropper tried to intercept the communication. The BB84 protocol proposed by Bennett and Brassard was one of the first QKD protocols ever and remains among the most widely known. BB84 protocol utilizes quantum states to encode information, and its security is derived from the property of measurement disturbing the system. If an eavesdropper attempts to observe the quantum bits that are sent, this causes observable errors in the communication. This work interfaces the BB84 protocol with a python simulation. Thus, we will explore the behavior of noise in the channel and how eavesdropper influences system performance. The analysis is based on key metrics such as QBER and key generation rate, which help in understanding both the security and efficiency of the protocol.

II. Literature review

Quantum cryptography has gained attention since the original BB84 protocol was proposed. The first application of quantum mechanics was shown by Bennett and Brassard [1] when they proved that the laws of quantum mechanics could be used for securely distributing cryptographic keys with detection of eavesdropping attempts. They laid the theoretical groundwork for quantum key distribution. Later investigations focused on practical implementation of quantum key distribution (QKD) systems employing optical fiber based communication. Researchers like Gisin and co-authors showed that quantum cryptography was achievable in practical situations, and they drew attention to difficulties caused by the loss of photons and channel noise. Recently, there has been a growing body of work to enhance the robustness and efficiency of QKD systems. Numerous simulation-based studies have been conducted to assess the performance of QKD protocols in terms of noise, noisy detectors, and dishonest detection attacks. They are part of efforts to help researchers understand how quantum cryptographic systems would work in real-world communication conditions. This work is based on prior studies that simulated the BB84 protocol to investigate the effects of noise and eavesdropping on them secure key generation.

III. BB84 Protocol Overview

The BB84 protocol enables two users, traditionally named Alice and Bob, to generate a shared secret key using quantum bits. The protocol uses two different measurement bases:

- Rectilinear basis (+)
- Diagonal basis (×)

Each basis represents binary values through different photon polarization states. The protocol continues as follows:

1. Alice produces a string of bits at random.
2. Alice randomly chooses her basis for encoding each bit.
3. Encoded photons are sent to Bob over a quantum channel.
4. Bob randomly choose a basis to measure each photon.

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5. Alice and Bob publicly compare the bases that they chose.
6. Variance bits using different bases are not taken into account.
7. Out of the remaining bits, a small subset is revealed to estimate Quantum Bit Error.
8. If the error rate is within acceptable limits, the remaining bits form the shared secret key.

As soon as an eavesdropper tries to measure the photons, they will leave some disturbance that will increase the error rate and eventually allow Alice and Bob to notice someone is trying attack.

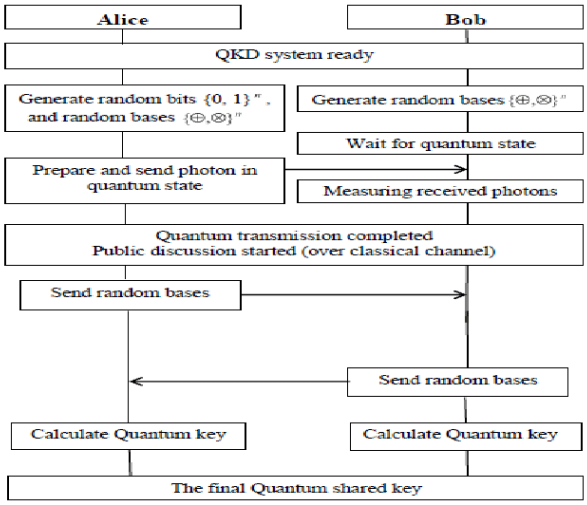

Fig. 1. Flow Diagram of BB84 Protocol

IV. System Model

The system model for this study includes three entities:

- Alice – the sender of quantum bits
- Bob – the receiver of quantum bits
- Eve – a potential eavesdropper

Alice sends photons to Bob through a quantum channel. Eve may intercept the transmission and attempt to measure the photons. A classical channel is used for basis comparison between Alice and Bob.

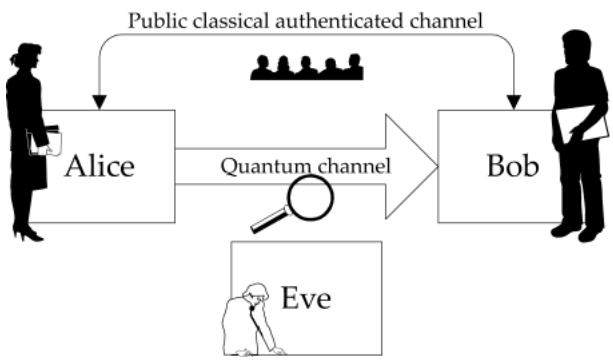

Fig. 2. System Architecture showing Alice, Bob and Eve.

V. Implementation

Python is used to simulation the BB84 protocol Random function is used to generate bit sequences and bases. Comparing the encoding bases to the measurement bases simulates transmission of photons. Noise exists in the system with a configurable probability to mimic real world scenarios. An intercept-resend attack is also modeled to emulate eavesdropping.

A. *Algorithm 1*

Input: Number of bits (n), noise probability (p), Eve flag
Output: QBER and final key

1) Generate random bit sequence
2) Generate random bases for Alice
3) Encode Photons
4) If Eve present → measure and resend
5) Bob measures using random bases
6) Introduce Noise
7) Compare Bases
8) Compute Qber

B. *Mathematical Model*

The Quantum Bit Error Rate is defined as:
QBER = (Number of incorrect bits) / (Total matched bits)

If Eve is present, additional errors occur due to incorrect basis selection, increasing QBER.

C. *Experimental Setup*

The simulation was performed using the parameters listed in Table I.

Table I

Parameters	Value
Number of bits(n)	1000
Number of trials	30
Noise range	0 – 0.2
Noise step size	0.025
Attack model	Intercept-resend

D. *Key Rate Calculation*

In addition to QBER, the performance of the BB84 protocol is evaluated using the secret key generation rate. The key rate represents the fraction of secure bits that can be extracted after error detection.
The key rate is computed using the relation:

$$\text{Key Rate} = 1 - 2 \times H(\text{QBER})$$

where H(QBER) represents the binary entropy function defined as:

$$H(x) = -x \log_2(x) - (1 - x) \log_2(1 - x)$$

This metric provides insight into how efficiently secure keys can be generated. As the QBER increases, the key rate decreases, indicating reduced reliability of secure communication.

The complete implementation of the BB84 protocol simulation is available in a public repository [5].

The core implementation of the BB84 protocol simulation is shown below.

```
def run_bb84(n, noise_probability=0.0, eve=False):

    # Alice
    bits_a = [random.randint(0,1)
              for _ in range(n)]
    basis_a = [random.choice(['+','x'])
              for _ in range(n)]

    # Bob
    basis_b = [random.choice(['+','x'])
              for _ in range(n)]
    bits_b = []


    # Transmission (with optional Eve)
    for i in range(n):
        if eve:
            eve_basis = random.choice(['+','x'])
            eve_bit = bits_a[i]
                if eve_basis == basis_a[i]
                else random.randint(0,1)
            bob_bit = eve_bit
                if basis_b[i] == eve_basis
                else random.randint(0,1)
        else:
            bob_bit = bits_a[i]
                if basis_b[i] == basis_a[i]
                else random.randint(0,1)

        bits_b.append(bob_bit)

    # Channel Noise
    for i in range(n):
        if random.random() < noise_probability:
            bits_b[i] = 1 - bits_b[i]

    # Key Sifting
    key_a = [bits_a[i]
             for i in range(n)
                if basis_a[i] == basis_b[i]]
    key_b = [bits_b[i]
            for i in range(n)
                if basis_a[i] == basis_b[i]]

    # QBER
    errors = sum(1 for i in range(len(key_a))
                if key_a[i] != key_b[i])
    qber = errors / len(key_a)
                if len(key_a) > 0 else 0

    # Key Rate
    correct_bits = sum(1
                        for i in range(len(key_a))
                        if key_a[i] == key_b[i])
    key_rate = correct_bits / n

    return qber, key_rate
```

Algorithm 2. Python implementation of the BB84 protocol with noise and intercept-resend attack.

The above code simulates the BB84 protocol by modeling Alice, Bob, and an optional eavesdropper. It includes channel noise, basis reconciliation, QBER computation, and key rate estimation for performance evaluation.

VI. Results

The simulation was conducted for multiple noise probabilities to evaluate the impact of channel noise on the Quantum Bit Error Rate (QBER) and the secret key generation rate. Two scenarios were analyzed:

1. Communication without an eavesdropper
2. Communication with an intercept-resend attack

As can be seen from the results, QBER gradually increases with the degradation of noise level because of the random disturbance in their communication channel. With no eavesdropper, the error rate is mostly low and only due to environmental noise effects. When an eavesdropper is in the communication channel, the error rate skyrockets from interference associated with rogue measurement basis.

Fig. 4 shows the dependence of QBER on noise probability without an eavesdropper. As shown on the graph — rate of error gradually getting higher

Fig. 5 shows a joint comparison of the QBER for both scenarios. The graph clearly indicates that given the same noise level, the QBER is always higher when eavesdropper exists than for normal communication. This discrepancy underscores the BB84 protocol's successful revelation of potential eavesdropping due to higher error rates.

Fig. 6 depicts the key rate as a function of noise probability. And as per the condition of obtaining the key rate, we see that the signal to noise ratio increases. A high Key Rate means converting the secure communication efficient. If QBER is not too high (less than the value of 25%) then key rate decreases slightly with increase in QBER, but after certain amount of QBER it approaches zero.

This behavior reflects the security versus efficiency trade-off of quantum key distribution systems. Increased noise levels and eavesdropping will both make attacks easier to detect, while at the same time reducing the amount of key bit clips that can be used, thus subsequently limiting secure communication.

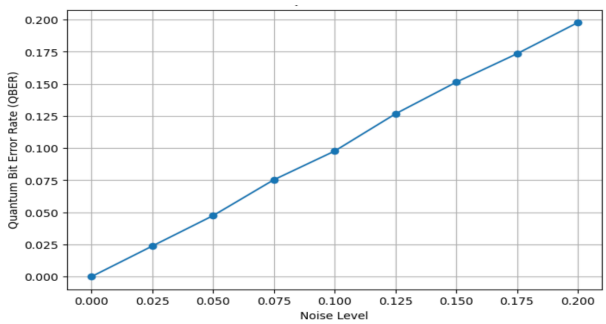

Fig. 3. QBER vs Noise Probability(No Eve)

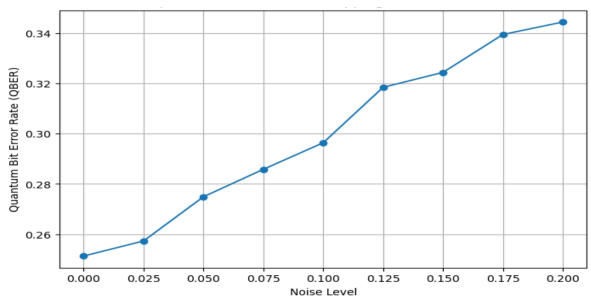

Fig. 4. QBER vs Noise Probability(With Eve)

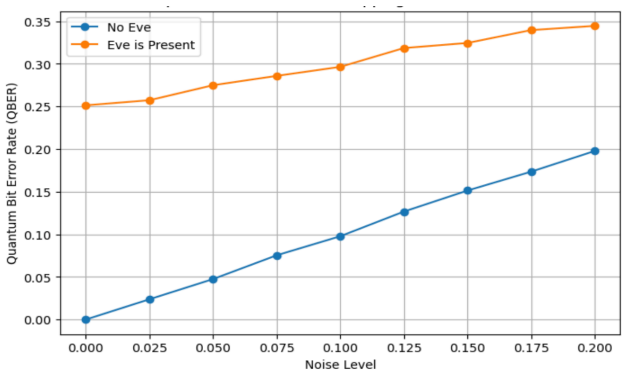

Fig. 5. QBER vs Noise Probability(Combined Graph)

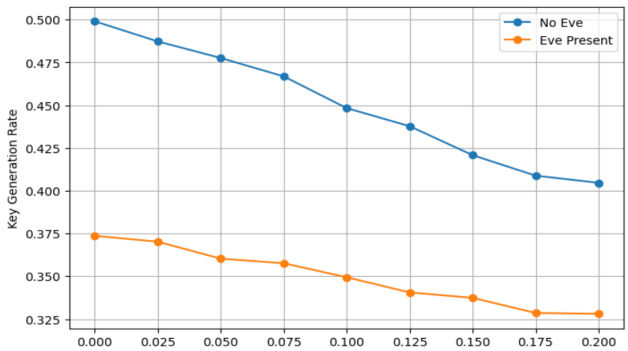

Fig. 6. Key Generation Rate vs Noise Level

This validates the robustness of the BB84 protocol in detecting eavesdropping through observable error patterns.

VII. Discussion

The simulation results clearly verity the principle of BB84 protocol. Finally, the increase in QBER due to an eavesdropper verifies that any measurement on a quantum system disturbs its state. This prompts errors in received data, detectable to the communicating parties. You can see the very difference in results between normal communication and communication under attack. This demonstrates a powerful advantage of quantum cryptography over its classical methods, where you might not always see eavesdropping. Another key observation is associated with the coupling of QBER and key generation rate. When the error increases, the number of bits is reduced, which decrease the efficiency of the system. Another observation is that the maximum value of QBER when it starts getting tough to create a secure key lies around 11%. Such a threshold is relevant when assessing the security of the communication. The results confirm that the BB84 protocol offers a secure method for secure key distribution. However, maintaining low noise levels in the communication channel is essential to ensure both security and efficiency.

VIII. Conclusion

In this paper, we implemented a simulation for BB84 quantum key distribution protocol to observe some of its performance metrics under different noisy environment and influence of an eavesdropper. Results of the simulations show that, in case any unauthorized access is attempted, an increase in the Quantum Bit Error Rate indicates our protocol's ability to detect it.

The analysis also indicates the trade-off between error rate and key generation. The more noise there is, the less reliable that generated key will be, thus limiting our ability to communicate securely. Such results establish BB84 in a practical and efficient way of key distribution.

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